Experiment: Configuration of IP Routing using RIP

1. Aim

To configure Dynamic IP Routing using RIP (Routing Information Protocol) Version 2 on a network comprising two subnets, a switch, and a router using Cisco Packet Tracer.

2. Requirements (Apparatus)

- **Simulator:** Cisco Packet Tracer (Student or Instructor Version)
- **Router:** 1 unit (e.g., Cisco 1941 or 2901)
- **Switch:** 1 unit (e.g., Cisco 2960)
- **End Devices:** 2 units (PC0, PC1)
- **Cables:** Copper Straight-Through and Copper Cross-Over cables

3. Network Topology Diagram

- **Network A (Left):** PC0 connected to Switch, Switch connected to Router (Gig0/0).
- **Network B (Right):** PC1 connected directly to Router (Gig0/1).

4. IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC1	NIC	192.168.2.10	255.255.255.0	192.168.2.1
Router	Gig0/0	192.168.1.1	255.255.255.0	N/A
Router	Gig0/1	192.168.2.1	255.255.255.0	N/A

5. Step-by-Step Procedure

Step 1: Physical Connections

1. Connect **PC0** (FastEthernet0) to **Switch** (FastEthernet0/1) using a **Straight-Through Cable**.
2. Connect **Switch** (GigabitEthernet0/1) to **Router** (GigabitEthernet0/0) using a **Straight-Through Cable**.
3. Connect **PC1** (FastEthernet0) directly to **Router** (GigabitEthernet0/1) using a **Cross-Over Cable**.

Step 2: Configure End Devices (PCs)

1. **For PC0:**
   - Go to **Desktop** > **IP Configuration**.
   - Set IP Address: `192.168.1.10`
   - Set Gateway: `192.168.1.1`
2. **For PC1:**
   - Go to **Desktop** > **IP Configuration**.
   - Set IP Address: `192.168.2.10`
   - Set Gateway: `192.168.2.1`

Step 3: Configure Router Interfaces

We must assign IP addresses to the router interfaces to act as gateways for the PCs.

- Click on **Router** > **CLI** tab.
- Enter the following commands:

```
Router> enable
Router# configure terminal
! Configure Left Interface (Gateway for Network A)
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

! Configure Right Interface (Gateway for Network B)
Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Step 4: Configure RIP Routing

This is the core step to enable the RIP protocol.

- In the same global configuration mode (`Router(config)#`), enter:

```
! Enable RIP
Router(config)# router rip

! Set version to 2 (Classless routing)
Router(config-router)# version 2

! Disable auto-summary to allow specific subnet masks
Router(config-router)# no auto-summary

! Advertise the connected networks
Router(config-router)# network 192.168.1.0
Router(config-router)# network 192.168.2.0
```

```
! Exit and Save
Router(config-router)# end
Router# write memory
```

6. Observation and Output

Verification A: Verify IP Routing Table

Check if the router recognizes the networks.

- **Command:** `show ip route`
- **Expected Output:**

```
Router# show ip route
...
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
C    192.168.2.0/24 is directly connected, GigabitEthernet0/1
L    192.168.2.1/32 is directly connected, GigabitEthernet0/1
```